

CS 170 Programming Quiz 9

For this quiz you will write **one** method that **takes arguments** and **returns a value**. The methods may require you to:

- ✓ Correctly define a method that *takes* and *returns* **arrays or lists**
- ✓ Process arrays and lists with *loops*
- ✓ Write functions that use counting, accumulation, extreme values and linear search

There is no printing and you will not need **Scanners**, or do any input.

You may not access any notes (other than the Java Quick Reference below), files, or communicate with anyone in any way (including checking your email or looking up files on the Q: drive.) When you have finished and submitted the quiz, check to see that I have received it.

You can check your problem by using **Tools->Compile** to see if your code is syntactically correct. To **test your code**, use **Tools->Go**. You can **run** the program (in the console) by using **Tools->Run in Console**.

Processing Data with Loops, Arrays & Lists

The Javadoc documentation for the function you need to write will be provided inside the starter file, **PQ09.java**. When you use **Tools->Go** it will test the method. You will get partial credit for each additional part you complete. The first set of tests make sure you have the correct "skeleton". These are called "smoke tests" and are worth about one-third of your quiz grade. If the smoke tests do not pass, then none of the correctness tests can be checked.

When you do **Tools->Run in Console** it will run the **main()** method at the bottom of the class. Add any additional code you like to the **main()** method. It is not part of the test, but it must compile.

Make sure you can successfully run Tools->Go before you submit your exam. The testing done on your machine is submitted for your grade.

Variable and Constant Declarations

Type **Name** **Initial value**
/
/
/
int cansPerPack = 6;

final double CAN_VOLUME = 0.335;

Method Declaration

Modifiers **Return type** **Parameter type and name**
/
/
/
/
public static double cubeVolume(double sideLength)
{
 double volume = sideLength * sideLength * sideLength;
 return volume;
}

Exits method and returns result.

Conditional Statement

Condition
/
if (floor >= 13)
{
 actualFloor = floor - 1;
}
else if (floor >= 0) Second condition (optional)
{
 actualFloor = floor;
}
else
{
 System.out.println("Floor negative");
}
}

Executed when condition is true

Executed when all conditions are false (optional)

Loop Statements

Condition
/
while (balance < TARGET)
{
 year++;
 balance = balance * (1 + rate / 100);
}

Executed while condition is true

Initialization **Condition** **Update**
/
/
/
for (int i = 0; i < 10; i++)
{
 System.out.println(i);
}

Loop body executed at least once
do
{
 System.out.print("Enter a positive integer: ");
 input = in.nextInt();
}
while (input <= 0);

Set to a new element in each iteration

An array or collection
for (double value : values)
{
 sum = sum + value;
}

Executed for each element

String Operations

```
String s = "Hello";  
int n = s.length(); // 5  
char ch = s.charAt(1); // 'e'  
String t = s.substring(1, 4); // "ell"  
String u = s.toUpperCase(); // "HELLO"  
if (u.equals("HELLO")) ... // Use equals, not ==  
for (int i = 0; i < s.length(); i++)  
{  
    char ch = s.charAt(i);  
    Process ch  
}
```

Mathematical Operations

Math.pow(x, y)	Raising to a power x^y
Math.sqrt(x)	Square root $\sqrt{x}$
Math.log10(x)	Decimal log $\log_{10}(x)$
Math.abs(x)	Absolute value $ x $
Math.sin(x)	} Sine, cosine, tangent of x (x in radians)
Math.cos(x)	
Math.tan(x)	

Class Declaration

```
public class CashRegister  
{  
    private int itemCount;  
    private double totalPrice;  
}  
  
public void addItem(double price)  
{  
    itemCount++;  
    totalPrice = totalPrice + price;  
}  
...  
}
```

Instance variables

Method

Selected Operators and Their Precedence

(See Appendix B for the complete list.)

[]	Array element access
++ -- !	Increment, decrement, Boolean <i>not</i>
* / %	Multiplication, division, remainder
+ -	Addition, subtraction
< <= > >=	Comparisons
== !=	Equal, not equal
&&	Boolean <i>and</i>
	Boolean <i>or</i>
=	Assignment

Input

```
Scanner in = new Scanner(System.in);
// Can also use new Scanner(new File("input.txt"));

int n = in.nextInt();
double x = in.nextDouble();
String word = in.next();
String line = in.nextLine();

while (in.hasNextDouble())
{
    double x = in.nextDouble();
    Process x
}
```

Output

```
System.out.print("Enter a value: ");

System.out.println("Volume: " + volume);

System.out.printf("%-10s %10d %10.2f", name, qty, price);

PrintWriter out = new PrintWriter("output.txt");

out.close();
```

Does not advance to new line.

Use + to concatenate values.

Field width
Precision

Left-justified string **Integer** **Floating-point number**

Use print/println/printf to write output to file.

Remember to close output file.

Arrays

```
int[] numbers = new int[5];
int[] squares = { 0, 1, 4, 9, 16 };
int[][] magicSquare =
{
    { 16, 3, 2, 13},
    { 5, 10, 11, 8},
    { 9, 6, 7, 12},
    { 4, 15, 14, 1}
};

for (int i = 0; i < numbers.length; i++)
{
    numbers[i] = i * i;
}

for (int element : numbers)
{
    Process element
}

System.out.println(Arrays.toString(numbers));
// Prints [0, 1, 4, 9, 16]
```

Element type **Element type** **Length**

All elements are zero.

Array Lists

```
ArrayList<String> names = new ArrayList<String>();

names.add("Ann");
names.add("Cindy"); // [Ann, Cindy], names.size() is now 2

names.add(1, "Bob"); // [Ann, Bob, Cindy]
names.remove(2); // [Ann, Bob]
names.set(1, "Bill"); // [Ann, Bill]

String name = names.get(0); // Gets "Ann"
System.out.println(names); // Prints [Ann, Bill]
```

Initially empty

Use wrapper type, Integer, Double, etc., for primitive types.

Element type

Add elements to the end

Linked Lists, Sets, and Iterators

```
LinkedList<String> names = new LinkedList<String>();
names.add("Bob"); // Adds at end

ListIterator<String> iter = names.listIterator();
iter.add("Ann"); // Adds before current position

String name = iter.next(); // Returns "Ann"
iter.remove(); // Removes "Ann"

Set<String> names = new HashSet<String>();
names.add("Ann"); // Adds to set if not present
names.remove("Bob"); // Removes if present

Iterator<String> iter = names.iterator();
while (iter.hasNext())
{
    Process iter.next()
}
```

Maps

```
Map<String, Integer> scores = new HashMap<String, Integer>();

scores.put("Bob", 10);
Integer score = scores.get("Bob");

for (String key : scores.keySet())
{
    Process key and scores.get(key)
}
```

Key type **Value type**

Returns null if key not present